WEEK 5 OVERVIEW

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CEVE 543

2026-09-21

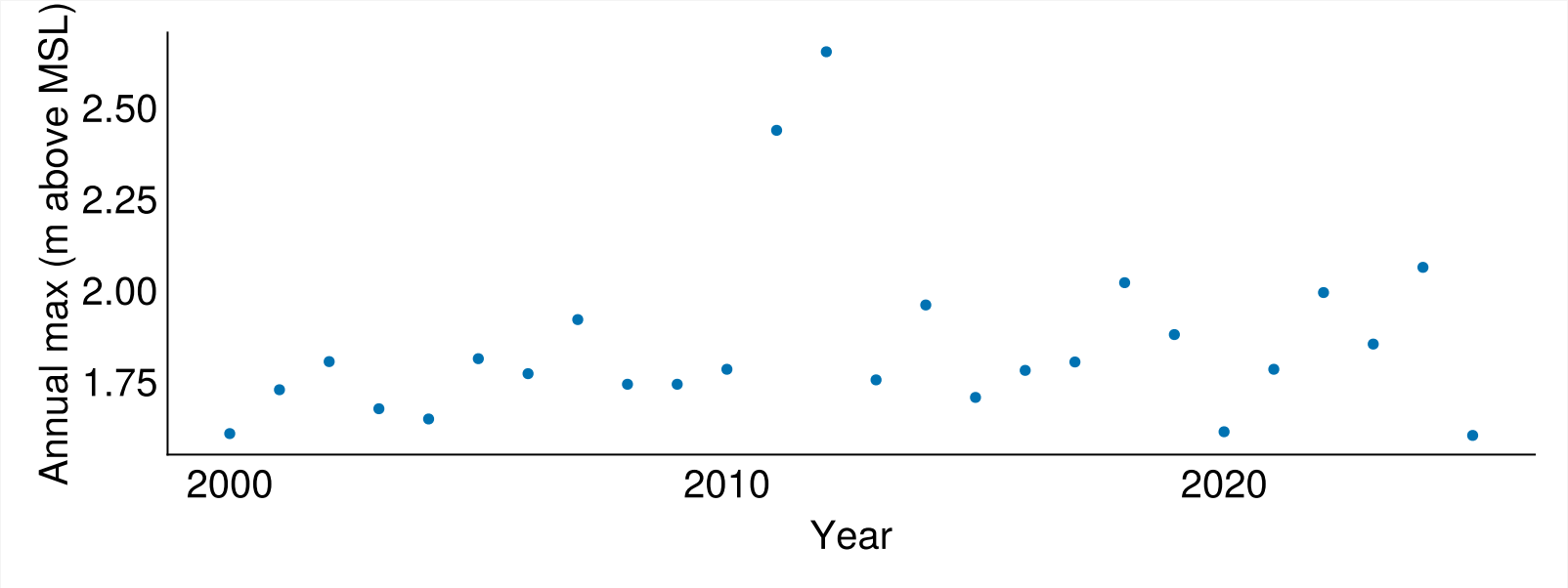


THIS WEEK

- **Monday:** peaks over threshold and the GPD (today)
- **Wednesday:** practice problems + Brays Bayou case study
- **Friday:** lab 5, remote (no in-person meeting)

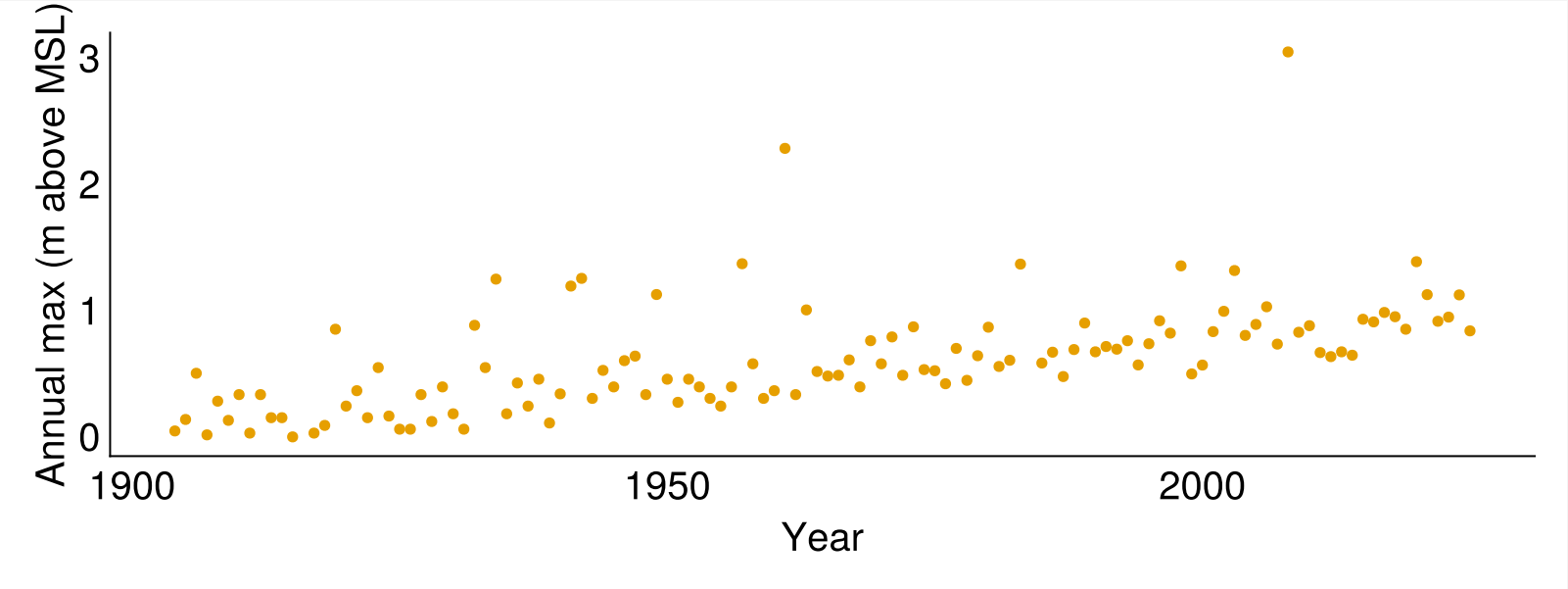
Lab 5 will be posted by Wednesday.

THE PROBLEM WITH BLOCK MAXIMA



New Haven, CT (NOAA 8465705). Sandy (2012) and Irene (2011) tower over everything else. Do we have 26 annual maxima, or 2 real extremes and 24 ordinary high tides?

SAME STORY, DIFFERENT COAST



Galveston Pier 21 (NOAA 8771450), 120 years of hourly data. Ike (2008, 3.0 m) and Carla (1961, 2.3 m) stand apart; everything else is under 1.4 m. Block

SAME STORY, DIFFERENT COAST

maxima treats the year with Ike and a quiet year the same way: one observation each.

WHY PEAKS OVER THRESHOLD?

Block maxima is **statistically wasteful** (Coles, §4.1). It reduces a year of sub-annual data to one number.

The HEC fishing analogy: block maxima keeps only the biggest fish each day. Peaks over threshold keeps every fish above the legal length.

Peaks over threshold uses **more of the data**, which means tighter confidence intervals on the same return level. The threshold is a modeling choice you can **diagnose** (mean residual life plot, parameter stability). “One year” is a convention nobody checks.

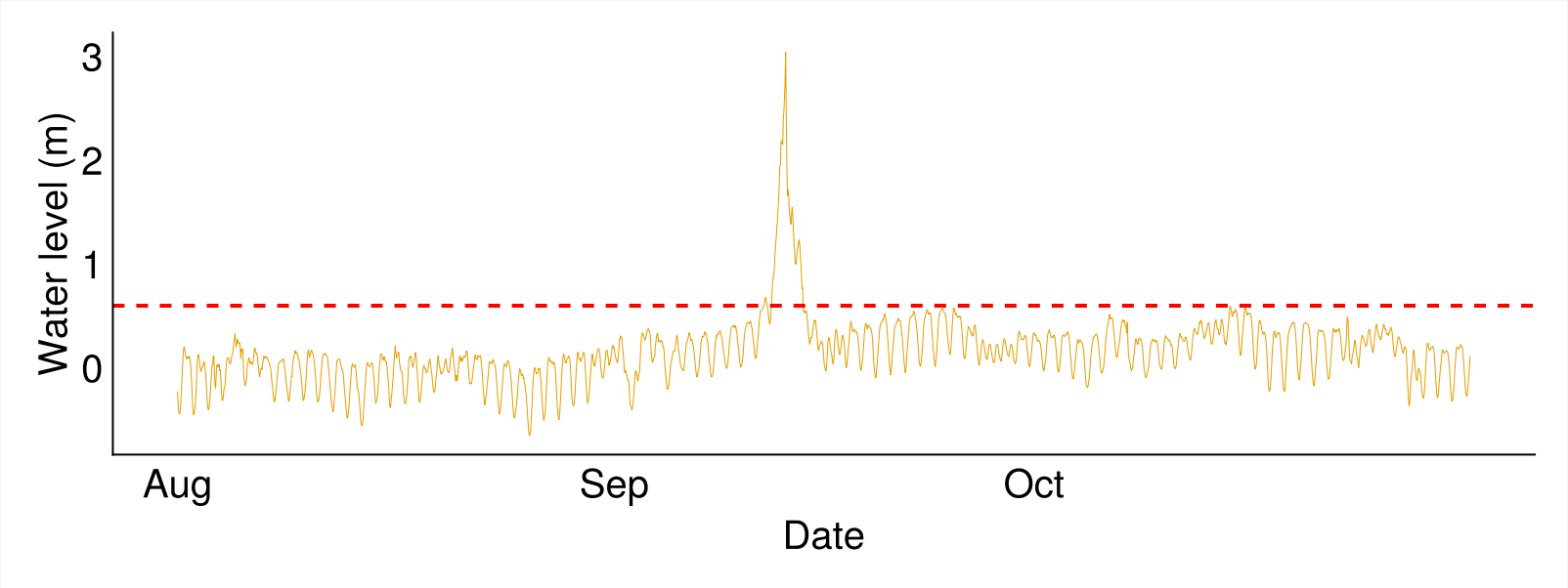
We need two ingredients:

1. **The distribution of exceedances** above the threshold (the GPD)

WHY PEAKS OVER THRESHOLD?

2. **How often exceedances happen** (a rate, events per year)

WHAT THE HOURLY RECORD LOOKS LIKE



Galveston, Aug–Oct 2008. The dashed line is a candidate threshold. Ike keeps the gauge above it for days: those are not independent exceedances.

TODAY'S PLAN

1. The GPD and compound-Poisson derivation (board)
2. Threshold diagnostics worked on the Coles rainfall and Galveston (slides)

The connection between the GEV and the GPD, the compound-Poisson structure, and the judgment calls in choosing a threshold.